

week3_tut

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Q1 & Q2

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.4.4      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.0
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
d = read.csv (file= "https://ckan0.cf.opendata.inter.prod-toronto.ca/dataset/21c83b32-d5a8-4106-a54f-010dbe49f6f2/resource/55d58477-50f5-490c-8da8-5953e3b26ca4/download/daily-shelter-overnight-service-occupancy-capacity-2022.csv")
```

##Q3

```
new_data = d %>% filter(PROGRAM_AREA=="Temporary Refugee Response" & SECTOR=="Families")
```

##Q4

```
new_data = new_data %>% select(LOCATION_CITY, CAPACITY_ACTUAL_ROOM, OCCUPIED_ROOMS, UNOCCUPIED_ROOMS)
```

##Q5

```
summary(new_data %>% select(where(is.numeric)))
```

```
## CAPACITY_ACTUAL_ROOM OCCUPIED_ROOMS UNOCCUPIED_ROOMS
## Min. : 2 Min. : 2.0 Min. :0.0000
## 1st Qu.: 80 1st Qu.: 80.0 1st Qu.:0.0000
## Median :118 Median :117.0 Median :0.0000
## Mean :106 Mean :105.8 Mean :0.2481
## 3rd Qu.:132 3rd Qu.:132.0 3rd Qu.:0.0000
## Max. :149 Max. :149.0 Max. :5.0000
```

##Q6

```
output = new_data %>% group_by(LOCATION_CITY) %>% summarise(SUM_UNOCCUPIED_ROOMS = sum(U
NOCCUPIED_ROOMS))

library(knitr)
kable(output)
```

LOCATION_CITY	SUM_UNOCCUPIED_ROOMS
North York	88
Toronto	77

##Q7

```
new_data %>% mutate(new_var = OCCUPIED_ROOMS == ( new_data$CAPACITY_ACTUAL_ROOM - new_da
ta$UNOCCUPIED_ROOMS)) %>% summarize(sum_check=sum(new_var),N=n())
```

```
##      sum_check      N
## 1          665 665
```